Data Cleaning and Quality Improvement Report

DATA CLEANING PROCESS

The data cleaning process improved dataset quality by standardizing formats, removing duplicates and redundant fields, reducing missing values, and creating an analysis-ready dataset.

Step	Data Cleaning Activity	Key Action Performed	Purpose
1	Data Inspection	Reviewed dataset structure, columns, and data types	Understand data quality and identify issues
2	Date Standardization	Converted date fields into a consistent format	Ensure accurate time-based analysis
3	Missing Value Assessment	Identified columns with high missing values	Improve data completeness and reliability
4	Duplicate Review	Checked duplicate rows and applicant identifiers	Maintain data integrity
5	Data Consistency Check	Standardized text and categorical values	Improve reporting accuracy
6	Structural Review	Evaluated multi-stage tracking fields	Understand dataset structure
7	Remove Irrelevant Columns	Dropped address, timestamp, and admin fields	Reduce dataset complexity
8	Remove Duplicate Records	Eliminated repeated applicant records	Prevent inaccurate analysis
9	Remove Constant Columns	Removed columns with no variation	Retain only useful variables
10	Complete Academic Information	Standardized Degree Level, Term, and Year fields	Improve analytical value
11	Standardize Program Names	Cleaned and unified program entries	Enable consistent categorization
12	Remove Redundant Variables	Deleted unnecessary tracking fields	Improve dataset efficiency
13	Remove Duplicate Information	Eliminated overlapping intake-related data	Avoid redundancy
14	Remove Invalid Records	Deleted incomplete and invalid entries	Improve dataset reliability
15	Final Dataset Preparation	Exported cleaned and analysis-ready dataset	Support reporting and visualization

Supporting Documentation

The detailed documentation containing all data cleaning steps, rationale, findings, and transformations is available at the link below*:*

Link: Detailed   Data Cleaning Steps

CLEANED DATASET SPREADSHEET

The cleaned dataset file is attached below. It contains the final analysis-ready data obtained after completing all data cleaning and quality improvement procedures

Attached File link : De Paul 's Cleaned Data

DATA COMPARISION SUMMARY

The raw dataset initially contained a large volume of incomplete and non-essential information, making it difficult to derive meaningful insights. Through a systematic data cleaning process, the dataset was transformed into a structured, reliable, and analysis-ready asset.

This transformation significantly improved data quality by reducing missing values, removing redundant fields, and retaining only the most relevant information required for analysis. The result is a streamlined dataset that supports accurate reporting, visualization, and decision-making.

Quantitative Impact

Data Quality Metric	Before Cleaning	After Cleaning	Improvement
Total Records	7,543	4,981	Refined Dataset
Total Data Cells	603k	70k	Reduction
Total Columns	80	14	82.5% Reduction
Missing Values	371,657	2,222	99.40% Reduction
Duplicate Records	0	0	Maintained
Analysis Readiness	Low	High	Significant Improvement

Overview of Major Changes

- Dataset Simplification.
- Reduction in Missing Values.
- Record Refinement
- Improved Data Quality

KEY INSIGHTS

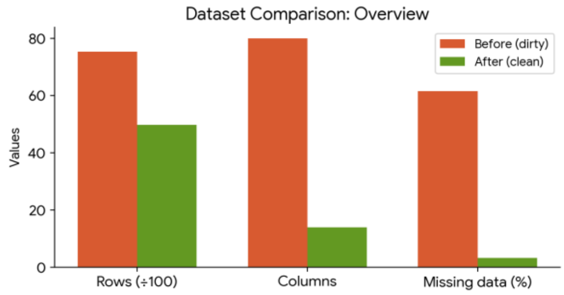
- Dataset complexity reduced by removing 82.5% of columns.
- Missing values reduced by approximately 99.40%.
- Data structure optimized for analysis and reporting.
- No duplicate records were identified in either dataset.
- Overall data completeness and reliability improved substantially.
- Cleaned dataset is better suited for visualization, dashboard creation, and business analysis.

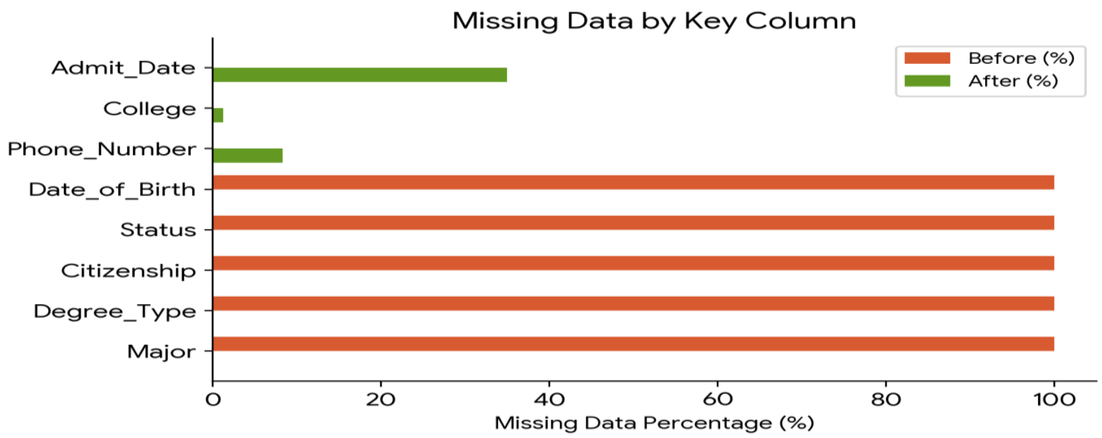
DATA VISUALIZATION

DePaul Dataset Comparison: Before and After Data Cleaning

AT A GLANCE

ROWS **7,543 → 4,981** 2,562 rows removed (34%)	COLUMNS **80 → 14** 66 columns dropped (83%)
MISSING CELLS **61.6% → 3.2%** From 371,657 to 2,222 nulls	TOTAL DATA CELLS **603K → 70K** Lean, focused dataset





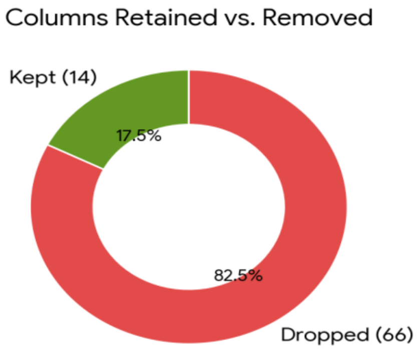
MISSING DATA BY KEY COLUMN

WHAT WAS FIXED

BEFORE (DIRTY)
80 columns — many redundant
14 duplicate column groups (e.g. -2, -3 suffixes)
Columns with 100% missing data (Degree_Type, Major, Status, Citizenship…)
7,543 total rows
Dates stored as raw timestamps (e.g. 1760000000000)
Phone numbers in inconsistent formats (e.g. 9.20E+11)
Country names inconsistent (United states vs United States)
Application source had inconsistent casing (STUDY GROUP)

AFTER (CLEAN)
14 clean, purposeful columns
All duplicates removed
Fully-empty columns dropped entirely
4,981 valid rows retained
Dates in readable ISO format (e.g. 2024-03-25)
Phone numbers standardised with + prefix
Country names consistently capitalised
Source values consistently cased (Study Group)

COLUMNS RETAINED VS. REMOVED



SPECIFIC CORRECTIONS MADE

Recieved_At → timestamp ➔ **Admit_Date → 2024-03-25 (readable date)**
Degree_Type (100% empty) ➔ **Degree_Level with 15 distinct, clean values (MS, MBA, MA…)**
Major + Major_1st_Choice (100% empty) ➔ **Program (populated, meaningful)**
Intake (empty) + multiple duplicate Intake-2 ➔ **Term + Year (split cleanly)**
United states / "Hong kong s.a.r." ➔ **United States (standardised casing)**
STUDY GROUP / Non Study Group ➔ **Study Group / Non Study Group (consistent)**
9.20E+11 (phone in scientific notation) ➔ **+91 XXXXX XXXXX (normalised format)**

Entire Folder LINK : Week_2 Deliverable

Short Summary Report

Overview

This project focused on improving the quality, reliability, and usability of the DePaul Admissions dataset through a structured data cleaning process. The raw dataset contained 7,543 records and 80 columns, with a large number of missing values, redundant tracking fields, duplicate applicant entries, and inconsistent formatting. These issues limited the dataset's effectiveness for analysis and reporting.

A comprehensive data cleaning process was performed, including data inspection, missing value assessment, date standardization, duplicate record removal, text standardization, elimination of redundant variables, and removal of irrelevant administrative fields. The dataset was then refined into a clean, structured, and analysis-ready format.

Impact on Dataset Reliability and Usability

The cleaning process significantly improved the overall quality of the dataset:

- Reduced columns from **80 to 14** by removing irrelevant and redundant variables.
- Reduced records from **7,543 to 4,981** by eliminating duplicate and invalid entries.
- Reduced missing values from **371,657 to 2,222**, achieving a **99.40% improvement in data completeness**.
- Standardized date fields, program names, and categorical values to improve consistency.
- Removed duplicate student records and unnecessary tracking information.
- Improved accuracy, reliability, and readability of the dataset.

These improvements resulted in a dataset that is more suitable for reporting, visualization, and analytical decision-making.

Initial Insights

The cleaning process revealed that a significant portion of the original dataset consisted of administrative tracking fields, duplicate applicant records, and columns with extremely high levels of missing data. By removing these issues, the final dataset became more focused on meaningful admissions-related information.

The cleaned dataset now supports:

- Accurate analysis of applicant trends and admission patterns.
- Reliable segmentation by program, degree level, term, and year.
- Improved dashboard and visualization development.
- More trustworthy business insights and decision-making.

Conclusion

The data cleaning process successfully transformed the DePaul Admissions dataset from a complex and incomplete dataset into a high-quality analytical resource. The final dataset is reliable, consistent, and ready for further analysis, providing a strong foundation for generating meaningful insights and supporting data-driven decisions.